

Proposed Solution Template

Project Design Phase

Project Name	Automated Network Request Management in ServiceNow
Phase	Phase 3: Project Design
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Platform / Domain	ServiceNow / IT Service Management (ITSM)
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Proposed Solution — Summary Table

1. Problem Statement

Enterprise network operations suffer from inefficiencies due to manual, ad-hoc request intake through emails, chats, and spreadsheets. This unstructured approach causes prolonged fulfillment cycles (high MTTR), zero request visibility for stakeholders, compliance audit risks from undocumented changes, manual SLA tracking, and frequent security vulnerabilities from unvetted infrastructure modifications.

2. Idea / Solution Description

Develop an automated, centralized Network Request Management system using the ServiceNow Service Catalog. Requesters submit technical details using dynamic, regex-validated forms. Multi-stage workflows configured in Flow Designer seamlessly route requests through Line Managers and Network Security teams for governance sign-off. Upon final approval, fulfillment tasks auto-route to the Network Engineering queue with active SLA monitoring and event-driven notifications at every lifecycle stage.

3. Novelty / Uniqueness

The solution uniquely unifies dynamic, client-side input validation (regex-checked IP/VLAN/port constraints) with complex, risk-based conditional routing within a single ServiceNow PaaS environment — without requiring any external tools, custom code deployments, or middleware integrations. All automation, governance, and audit logging are native to the platform.

4. Target Users

IT Staff and Business Users (requesters), Line Managers and Security Team Leads (approvers), Network Operations Engineers (fulfillment operators), and IT Administrators (oversight and reporting).

5. Key Features

8 structured Service Catalog items with dynamic, validated forms. Multi-stage Flow Designer approval workflows with conditional routing. Automated email notifications at every stage. Automated sc_task fulfillment task generation and assignment. Real-time SLA tracking with escalation at 50/75/100% thresholds. Role-based access control via ACLs. Complete immutable audit trail.

6. Technology Platform

ServiceNow Enterprise PaaS Cloud — Service Catalog, Flow Designer, UI Policies, Client Scripts, Notification Engine, SLA Management, Reporting & Dashboards, Update Sets for deployment.

7. Expected Outcomes

Elimination of email-based request intake. Reduction in MTTR. 100% audit coverage for all network change requests. Improved SLA compliance. Standardized, reportable request data. Scalable infrastructure to support growing organizational request volumes.

8. Future Roadmap

Integration with Cisco DNA Center and SolarWinds for automated network configuration execution. Virtual Agent chatbot for conversational request submission. Extended catalog covering cloud resource provisioning and server infrastructure requests. Real-time analytics dashboard for management reporting.